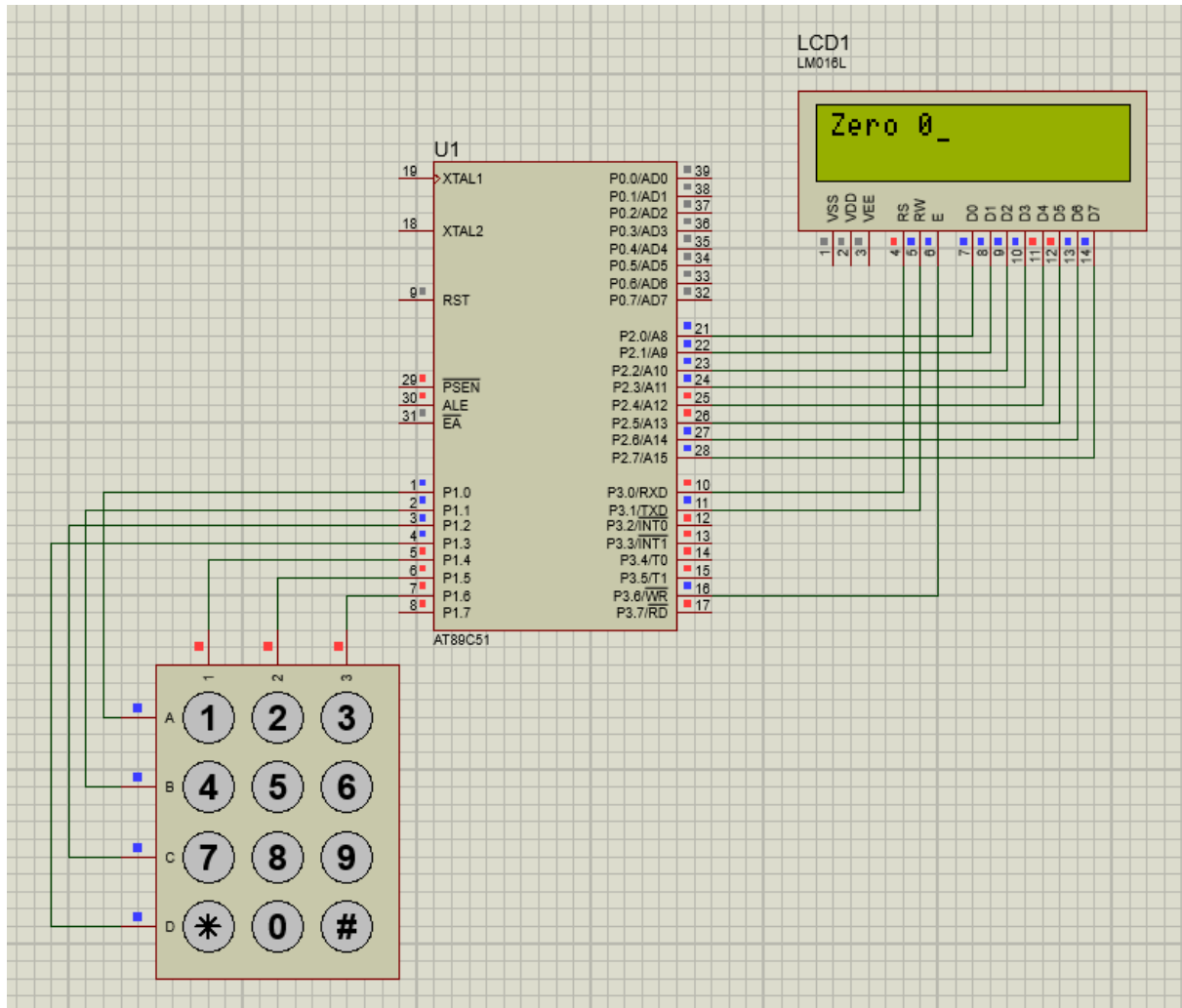
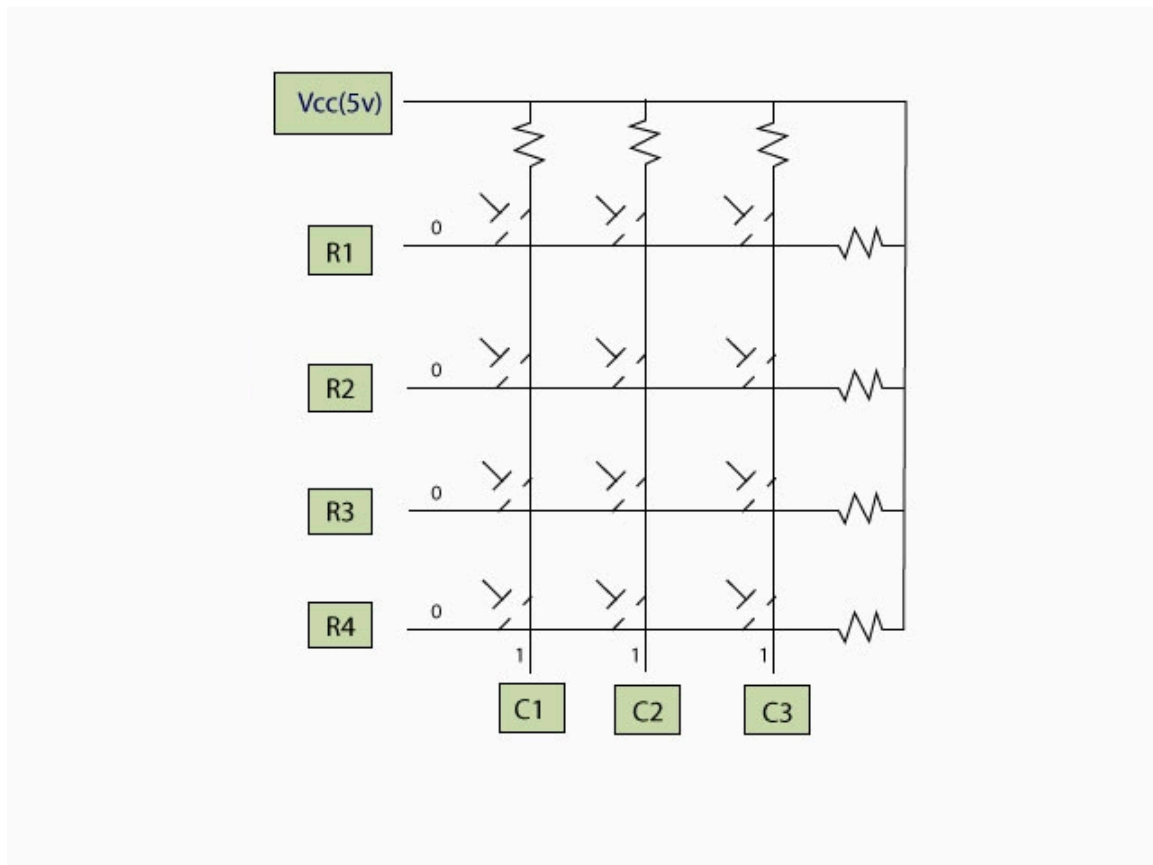


7 Keypad Input - LCD Output

Aim: To interface a keypad with the 8051 microcontroller & display the pressed key on the LCD.





Apparatus Required:

- 8051 microcontroller
- Keypad
- LCD
- Breadboard
- Connecting wires
- Power supply (5V)

Theory:

- A keypad is an I/P device consisting of multiple keys arranged in a matrix format (rows & columns).
- It is used to provide user I/P to the microcontroller.
- In a matrix keypad, keys are connected in such a way that pressing a key connects a specific row & column.
- The microcontroller scans the keypad by sending signals to rows & checking columns to detect which key is pressed.

- When no key is pressed, the I/P lines remain at logic HIGH due to pull-up resistors.
- When a key is pressed, the corresponding line becomes LOW.
- In this experiment, the detected key value is displayed on an LCD, demonstrating both I/P handling & O/P display interfacing.

Working:

- The microcontroller continuously scans the keypad by sending signals to its rows & checking the column lines for any key press.
- When a key is pressed, a connection is established b/w a specific row & column.
- This connection allows the microcontroller to detect the pressed key.
- The corresponding key value is then identified & processed.
- Finally, this detected value is sent to the LCD, where it is displayed as a character on the screen.

Keypad Mapping:

	Col 1	Col 2	Col 3
Row 1	1	2	3
Row 2	4	5	6
Row 3	7	8	9
Row 4	* (11)	0 (10)	# (12)

LCD Display Output:

I/P Value	LCD Display O/P
1	one 1
2	two 2
3	three 3
4	four 4
5	five 5
6	Six 6
7	Seven 7
8	Eight 8
9	Nine 9
10	Zero 0

11	*
12	#

Program Code:

```
#include <reg51.h>

#define port P3
#define dataport P2
#define key P1

sbit rs = port^0;
sbit rw = port^1;
sbit en = port^6;

sbit col1 = key^4;
sbit col2 = key^5;
sbit col3 = key^6;

// Standard delay function
void delay(unsigned int msec) {
    int i, j;
    for (i = 0; i < msec; i++)
        for (j = 0; j < 1275; j++);
}

// LCD Command function
void lcdcmd(unsigned char item) {
    dataport = item;
    rs = 0; rw = 0; en = 1;
    delay(1); en = 0;
}

// LCD Data function
void lcddata(unsigned char item) {
    dataport = item;
    rs = 1; rw = 0; en = 1;
    delay(1); en = 0;
}

// Initialize the LCD once at startup
void lcd_init() {
```

```

    lcdcmd(0x38); // 8-bit mode, 2 lines
    lcdcmd(0x0E); // Display ON, Cursor ON
    lcdcmd(0x01); // Clear Display
    lcdcmd(0x06); // Auto-increment cursor
}

// Clear screen and print the new string
void lcd_print(char *str) {
    lcdcmd(0x01); // Clear display so text doesn't overlap
    lcdcmd(0x80); // Force cursor to start of line 1

    while (*str != '\0') {
        lcddata(*str);
        str++;
        delay(10);
    }
}

void main() {
    // 1. A clean Lookup Table that mimics the physical keypad
    char *keypad_matrix[4][3] = {
        {"One 1", "Two 2", "Three 3"},
        {"Four 4", "Five 5", "Six 6"},
        {"Seven 7", "Eight 8", "Nine 9"},
        {"*", "Zero 0", "#"}
    };

    // 2. Hex values to ground one row at a time (Row 1 to 4)
    unsigned char row_mask[4] = {0xFE, 0xFD, 0xFB, 0xF7};
    int r;

    lcd_init();

    while (1) {
        key = 0xF0; // Set all Rows (P1.0-P1.3) LOW, and Cols (P1.
4-P1.6) HIGH

        // Check if ANY button is pressed
        if (col1 == 0 || col2 == 0 || col3 == 0) {
            delay(20); // Debounce: Wait 20ms to ignore physical b
utton noise

            // Loop through the 4 rows to find exactly which butto

```

```

n it is
    for (r = 0; r < 4; r++) {
        key = row_mask[r]; // Ground current row

        // If a column drops to 0, match it with the row
        (r) to print the text
        if (col1 == 0) { lcd_print(keypad_matrix[r][0]); w
hile(col1==0); }
        if (col2 == 0) { lcd_print(keypad_matrix[r][1]); w
hile(col2==0); }
        if (col3 == 0) { lcd_print(keypad_matrix[r][2]); w
hile(col3==0); }
    }
}
}
}

```

Result: The pressed key was successfully displayed on the LCD.

Precautions:

- Avoid floating I/Ps.
- Use pull-up resistors.

SOURCE: <https://www.engineersgarage.com/how-to-interface-keypad-with-8051-microcontroller-at89c51-part-13-45>